



REPORT

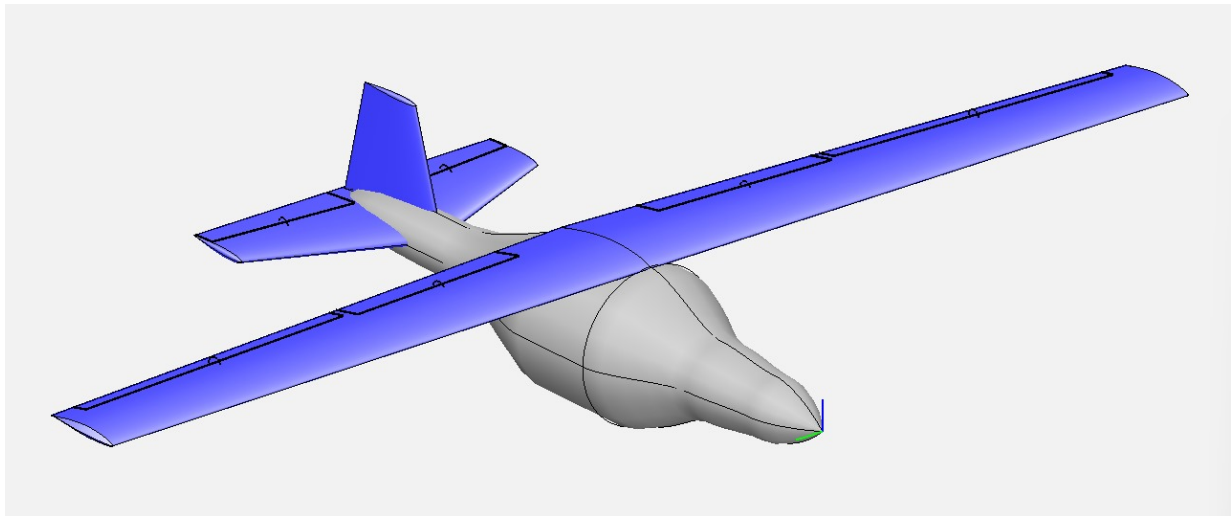
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Project 1 - Design of a 2-Seater General Aviation eAircraft

Fundamentals of Flight



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1 Introduction

The transition to decarbonization concerns the world of General Aviation as well, resulting in a growing interest in converting already existing aircrafts to electrical propulsion and designing completely new aircrafts. The aim of this project is indeed to conceptually design a 2-Seater General Aviation electrical aircraft. The starting point of the designed aircraft is the already existing Pipistrel Velis Electro.

1.1 Requirements

The design process of the 2-seater General Aviation eAircraft was conducted considering the following requirements:

1. Cruise altitude: 12000 ft (approximately 3600 m)
2. Cruise velocity range: 80-100 KCAS
3. Target Maximum Take-Off Weight Range: 500-600 kg
4. Payload (Pilot + Passenger): 180 kg
5. Endurance: 90min

Furthermore, for what concerns the Power system, the Pipistrel E-811 EASA Type-Certified was used.

2 Method

2.1 Constraints

The first step of the design process concerns the calculation of the constraints of the aircraft. This was done by plotting the constraint curves for the wing loading W/S and thrust-to-weight T/W ratio. The used equations are listed below, for the parameters check Table 7 in *Appendix A*.

1. T/W for a desired Take-Off ground run distance:

$$\frac{T}{W} = \frac{1.21}{g\rho C_{L_{max}} S_G} \left(\frac{W}{S} \right) + \frac{0.605}{C_{L_{max}}} (C_{D_{TO}} - \mu C_{L_{TO}}) + \mu \quad (1)$$

2. T/W for a Desired Rate of Climb:

$$\frac{T}{W} = \frac{V_v}{V_\infty} + \frac{q}{(W/S)} C_{D_{min}} + \frac{k}{q} \left(\frac{W}{S} \right) \quad (2)$$

3. T/W for a Desired Maximum Angle of Climb:

$$\frac{T}{W} = \sin\gamma + \frac{1}{LD_{max}} = \sin\gamma + \sqrt{4kC_{D_{min}}} \quad (3)$$

4. T/W for a Level Constant Velocity Turn:

$$\frac{T}{W} = q \left[\frac{C_{D_{min}}}{(W/S)} + k \left(\frac{n}{q} \right)^2 \left(\frac{W}{S} \right) \right] \quad (4)$$

5. T/W for a Climbing Constant Velocity Turn:

$$\frac{T}{W} = q \left[\frac{C_{D_{min}}}{(W/S)} + k \left(\frac{n}{q} \right)^2 \left(\frac{W}{S} \right) \right] + \frac{P_S}{V_\infty} \quad (5)$$

6. T/W for a Desired Cruise Airspeed:

$$\frac{T}{W} = q C_{D_{min}} \left(\frac{1}{W/S} \right) + \left(\frac{1}{q} \right) \left(\frac{W}{S} \right) \quad (6)$$

7. T/W for a Desired Service Ceiling:

$$\frac{T}{W} = \frac{1.667}{V_Y} + \frac{q}{(W/S)} C_{D_{min}} + \frac{k}{q} \left(\frac{W}{S} \right) \quad (7)$$

8. T/W for a Desired Cruise Lift-to-Drag Ratio:

$$\frac{T}{W} = \frac{1}{L/D} \quad (8)$$

9. W/S for a Target Total Landing Distance:

$$S_{LDG} = 19.08 h_{obst} + \left[0.007923 + 1.556 \tau \sqrt{\frac{A}{(W/S)}} + \frac{1.21}{g \left[\frac{0.605}{C_{L_{max}}} (C_{D_{LDG}} - \mu C_{L_{LDG}}) + \mu - \frac{T_{gr}}{W} \right]} \right] \frac{(W/S)}{A} \quad (9)$$

$$S_{LDG} = \left[0.01583 + 1.556 \tau \sqrt{\frac{A}{(W/S)}} + \frac{1.21}{g \left[\frac{0.605}{C_{L_{max0}}} (C_{D_{LDG}} - \mu C_{L_{LDG}}) + \mu - \frac{T_{gr}}{W} \right]} \right] \frac{(W/S)}{A} \quad (10)$$

The obtained graph (Figure 1) allowed to find the optimum design points in terms of thrust loading T/W and wing loading W/S .

In particular, the chosen point is characterized by the following wing loading and thrust loading:

$$\frac{W}{S} = 9.81 \text{ lb}_f \cdot \text{ft}^{-2} = 469.71 \text{ N} \cdot \text{m}^{-2} \quad \frac{T}{W} = 0.16$$

These values were considered in the following parts of the project, especially for what concerns the performance of the aircraft. The found wing loading also set the starting point for the estimation of the wing area: knowing the initial gross weight, of around 600 kg (approximately 1300 lb_f), the planform area is then equal to:

$$S = \frac{469.71 \text{ N} \cdot \text{m}^{-2}}{600 \text{ kg} \cdot 9.81 \text{ m} \cdot \text{s}^{-2}} = 12.53 \text{ m}^2 = 134.97 \text{ ft}^2 \quad (11)$$

This value was used for the wing geometry and aerodynamic calculations.

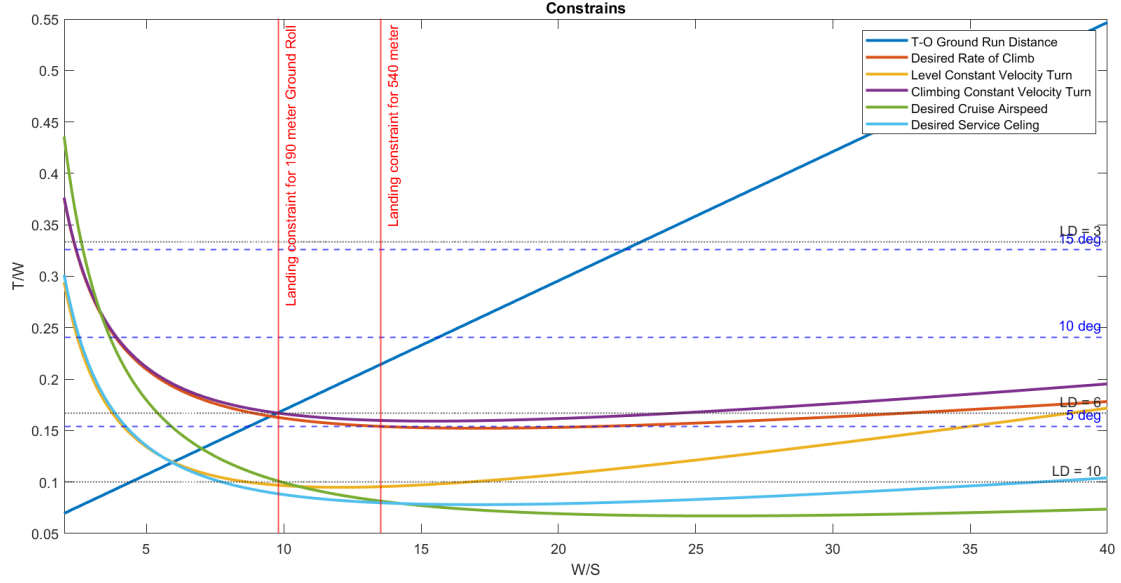


Figure 1: Constraint diagram (MATLAB).

2.2 Weight Analysis

For what concerns the weight analysis, the process consisted in the initial gross weight estimation, followed by the estimation of the weight of the different parts of the aircraft. The gross weight resulted 600kg. For the wings, a direct weight estimation was carried out, as they play a major role in the aerodynamic of the aircraft, whereas, for the other components, a statistical approach was carried out. For both approaches, it is the mass that was calculated, even if it is referred to as weight: it is indeed expressed in kilograms instead of Newtons. Here follow the used equations, usually applied for Cessna aircrafts, and the derived numerical results.

1. Wings; a direct weight estimation leads to calculating the weight of the sub-components of the wings, i.e. ribs, skin, web and spar caps. The flap length was assumed to be $\frac{1}{3}$ of the entire wing. The used formulas are:

$$W_{ribs} = N_{ribs} \rho_{ribs} \frac{A_{cell}(1 + \lambda^2)(t_{skin} + t_{skin_T})}{4} = 4.816 \text{ kg} \quad (12)$$

$$W_{skin} = \rho_{skin} \frac{b(t_{skin} + t_{skin_T})s_{cell}(1 + \lambda)}{8} = 32.561 \text{ kg} \quad (13)$$

$$W_{web} = \rho_{web} \frac{b(t_{web} + t_{web_T})h(1 + \lambda)}{8} = 6.139 \text{ kg} \quad (14)$$

$$W_{caps} = \rho_{caps} \frac{b(A_{cap} + A_{cap_T})}{2} = 19.156 \text{ kg} \quad (15)$$

where the material for all the parts is carbon fiber reinforced polymer so the density is $\rho = 1600 \text{ kg} \cdot \text{m}^{-3}$, the number of ribs is $N_{ribs} = 12$, the skin and web thickness is $t_{skin} = t_{web} = 0.005 \text{ m}$, the skin and web thickness at the tip is $t_{skin_T} = t_{web_T} = 0.001 \text{ m}$, the taper ratio is $\lambda = 0.7$, the cell area is $A_{cell} = 0.148 \text{ m}^2$, the wingspan is $b = 12.287 \text{ m}$, the cell arc length is $s_{cell} = 1.823 \text{ m}$, the structural depth at the mean geometric chord is $h = 0.245 \text{ m}$ and the spar cap area is $A_{cap} = 0.000475 \text{ m}^2$. All these values were obtained either from the analysis on the wing geometry that follows in the report or from the book General Aviation Aircraft

Design [1].

The sum of all these contributions give the total weight of the wings, excluding flaps.

$$W_{wings_{noflaps}} = W_{ribs} + W_{skin} + W_{web} + W_{caps} = 62.673 \text{ kg}$$

Considering the flaps, whose weight was estimated to be one fourth of the one just found, the total weight of the wings results:

$$W_{wings} = 78.342 \text{ kg}$$

2. Fuselage; the used expression predicts the weight of the fuselage shell and the associated internal structure, excluding furnishing and considering high-wing.

$$W_{fus} = 14.86W_0^{0.144} \left(\frac{l_{FS}}{R_{max}} \right)^{0.778} l_{FS}^{0.383} N_{OCC}^{0.455} = 104.455 \text{ kg} \quad (16)$$

where the gross weight $W_0 = 600 \text{ kg}$, the length of the fuselage is $l_{FS} = 21.3 \text{ ft} = 6.492 \text{ m}$, the fuselage max perimeter is $R_{max} = 15.46 \text{ ft} = 4.712 \text{ m}$ that derives from the choice of having a cockpit of maximum radius of 0.75 m , and the number of occupants is $N_{OCC} = 2$.

3. Landing gear; a fixed main landing gear was considered.

$$W_{lg} = 6.2 + 0.0143W_0 + 0.362W_l^{0.417} n_l^{0.950} L_m^{0.183} + 0.007157W_l^{0.749} n_z L_n^{0.788} = 23.667 \text{ kg} \quad (17)$$

where the design gross and landing weights are the same $W_0 = W_l = 600 \text{ kg}$, the ultimate landing factor is $n_l = 4$ and the length of the main landing gear shock strout is $L_m = 1 \text{ ft} = 0.3048 \text{ m}$.

4. Furnishing; the furnishing estimation includes seats, insulation, sound proofing, lighting and associated electric systems:

$$W_{furn} = 0.0412N_{OCC}^{1.145}W_0^{0.489} = 3.062 \text{ kg} \quad (18)$$

5. Windows; the windows weight was calculated by modeling the dimensions of the wind-shield ($4.25 \times 2.083 \times 0.0104 \text{ ft}^3$) and the side windows ($2.75 \times 1.33 \times 0.0104 \text{ ft}^3$) similar to those of a Cessna 172, using plexiglass, whose density is $33.7 \text{ kg} \cdot \text{ft}^{-3}$. The total weight results as:

$$W_{windows} = 5.682 \text{ kg}$$

6. Flight Control System; the formula for manual control system was used:

$$W_{ctrl} = 0.0168W_0 = 10.08 \text{ kg} \quad (19)$$

7. Avionics System; the following equation comprises the weight of the electronic navigation (NAV) and communication (COM) systems:

$$W_{av} = 2.11W_{UAV}^{0.933} = 1.144 \text{ kg} \quad (20)$$

8. Horizontal and vertical tail; for the horizontal tail:

$$W_{HT} = \frac{3.184W_0^{0.887}S_{HT}^{0.101}AR_{HT}^{0.138}}{174.04t_{HT_{max}}^{0.223}} = 9.655 \text{ kg} \quad (21)$$

For the vertical tail:

$$W_{VT} = (1 + 0.2F_{tail}) \frac{1.68W_0^{0.567}S_{VT}^{0.1249}AR_{VT}^{0.482}}{15.6t_{VT_{max}}^{0.747}(\cos\Lambda_{VT})^{0.882}} = 7.821 \text{ kg} \quad (22)$$

where the surface of the horizontal tail is $S_{HT} = 3.265 \text{ m}^2$, of the vertical tail is $S_{VT} = 2.247 \text{ m}^2$, the aspect ratio of the horizontal tail is $AR_{HT} = 3.5$, of the vertical tail is $AR_{VT} = 1.35$, the maximum root chord thickness of the horizontal tail is $t_{HT_{max}} = 0.155 \text{ m}$, of the vertical tail is $t_{VT_{max}} = 0.194 \text{ m}$ and the sweep angle at the quarter of the mean geometric chord is $\Lambda_{VT} = 0.507 \text{ rad} = 29.054^\circ$. The tail type is conventional.

9. Payload; the payload weight is part of the requirements and corresponds to the weight of the pilot and a passenger, thus resulting in:

$$W_{payload} = 180 \text{ kg}$$

10. Engine and power controller; this component was a given part of the aircraft, so its weight was not estimated but was taken from the manual of the given engine:

$$W_{eng} = 22.7 \text{ kg} \quad W_{pc} = 7 \text{ kg}$$

11. Cooling system; the cooling system was designed to use mineral oil as coolant liquid (density of $924.1 \text{ kg} \cdot \text{m}^{-3}$). It was estimated that 5.4 litres are needed to cool the batteries and 0.9 litres for the engine. Considering also the structure of the cooling system, whose weight is around 2.95% of the coolant weight, the total weight to cool down engine and batteries is:

$$W_{cool} = 5.993 \text{ kg}$$

12. Batteries; the estimation of the weight of the batteries, and, consequently, of the number needed has to go through the analysis of range and performance of the aircraft. The process can be found in the *Performance* section (3.2.1): it results in a required battery weight of $m_{batt} = 139.845 \text{ kg}$. Given that the weight of one of the chosen batteries is equal to 70 kg, the designed aircraft will include two batteries, for a total weight of 140 kg.

The total weight of the aircraft, summing up all the examined components, is the equal to:

$$W_{tot} = W_{maxTO} = 599.601 \text{ kg}$$

which actually corresponds to the initial gross weight estimation and to the maximum take-off weight.

The empty weight is then equal to the total weight without the payload, so:

$$W_{empty} = W_{tot} - W_{payload} = 419.601 \text{ kg}$$

2.3 Aircraft Configuration

The configuration and the placement of the aircraft's components are used to calculate the model center of gravity (CG). The position of its major components such as tail, wings, batteries, motor, power system, flight control and avionics systems, payload, landing gear, etc, were decided based on the datum point located at the leading edge of the wing, positioned at a distance of 2.34 m from the nose of the aircraft. With this datum reference, distances were measured to accurately determine each component's influence on the aircraft's overall CG. The specific values are gathered in table 8 in *Appendix B*.

The CG of the aircraft is then calculated by summing the mass·distance value of each component and dividing the result by the total weight of the aircraft, giving the x_{cg} of the aircraft from the leading edge of the wing:

$$x_{cg} = \frac{\sum(M \cdot x)}{M_{tot}} = -0.000691 \text{ m} \quad (23)$$

For what concerns the aerodynamic centre, it is further discussed in the *Longitudinal Stability and Control* section (Table 5 and Equation 76).

2.4 Wing Geometry

2.4.1 Airfoil Choice

For the selection of airfoil profiles, the three specific designs recommended in the reference text were considered. First, the NACA 23012, a classic airfoil with a high maximum lift coefficient ($C_{l_{max}}$), making it suitable for applications requiring strong lift performance at lower speeds. Next, the NACA 65₁412, an efficient laminar flow airfoil noted for its optimal lift-to-drag ratio, is often chosen in designs where maximizing aerodynamic efficiency is essential. Finally, the NLF 0414 F —part of the Natural Laminar Flow (NLF) series—is engineered with a pressure distribution that promotes an extended favorable pressure gradient. This characteristic helps maintaining a natural laminar boundary layer over a significant portion of the surface, reducing drag and enhancing fuel efficiency, especially at moderate speeds.

Initially, the airfoil profiles from the website *Airfoil Tools* [3] were obtained. Upon initial inspection, the NLF 0414 F appears to have a thicker profile, which may contribute to its stability and lift at lower speeds, while the NACA 65₁412 has a more streamlined shape, suggesting a design optimized to reduce drag and improve efficiency in cruise conditions.

For a more detailed analysis, we used the open-source software Xfoil [4], developed at MIT, to simulate each airfoil's performance at specific Reynolds numbers. Two primary Reynolds numbers were considered: 1.5×10^6 to simulate near-stall conditions, and 3×10^6 to represent steady flight at cruise levels. These simulations allowed the evaluation of each airfoil performance across a range of parameters.

The data collected were organized into a comparison table, which includes key performance metrics for each airfoil, such as the maximum lift coefficient ($C_{l_{max}}$) — higher values indicate superior lift potential — and the pitching moment coefficient at cruise (C_m), where lower values are generally preferable for stability and control. This structured comparison provided a basis for grading each airfoil in terms of lift, efficiency, and stability, enabling an informed selection of the optimal profile for our application.

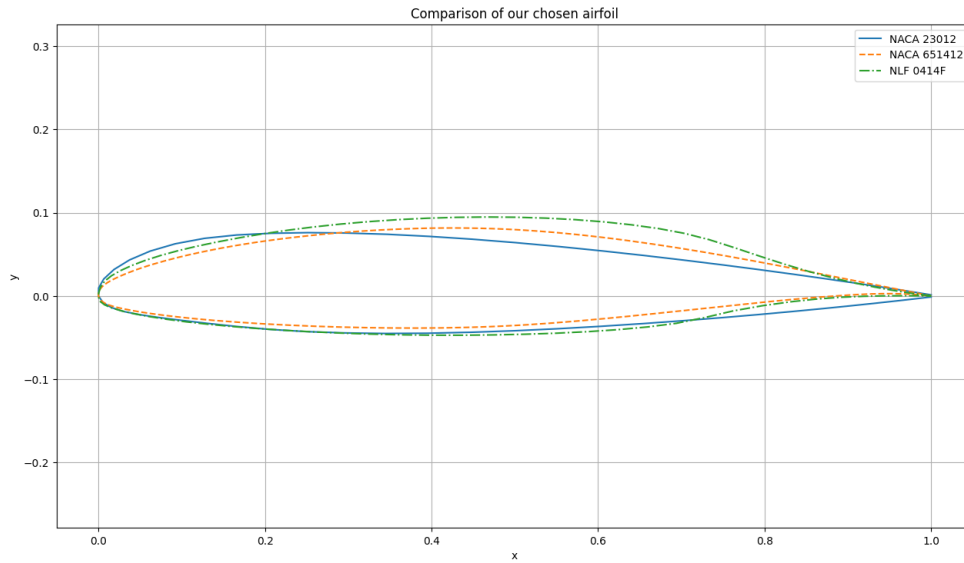


Figure 2: Plot of the 3 different airfoil profiles.

In the end, the cumulative grades for each airfoil revealed that the NACA 65₁412 outperformed the others, particularly due to its exceptional $(C_l/C_d)_{max}$ ratio, which indicates superior aerodynamic efficiency. This result confirmed the initial choice of the NACA 65₁412 a 12% thick airfoil designed for high lift-to-drag efficiency, essential for minimizing electrical energy consumption and enhancing performance at cruise speeds.

Given the design requirements for the two-seat electric airplane, the NACA 65₁412 offers the best balance between high aerodynamic efficiency and low drag, critical for maximizing the range and energy efficiency of an electric aircraft. Its ability to maintain stable, efficient flight at moderate speeds with low power consumption makes it the optimal choice for the aimed lightweight and energy-efficient design.

2.4.2 Wing Sizing

Following the selection of the airfoil, the 3D model was developed starting from the wing loading value derived from the constraint diagram (Equation 11).

Knowing the weight estimation, it was possible to calculate the wing area and from that, using the aspect ratio from the Pipistrel [5], also the wing span and the mean geometric chord were derived.

The chosen shape is trapezoidal, with a straight leading edge and a tapered trailing edge. This design was selected to minimize the drag caused by the vortexes at the wing tips and to simplify the calculations for the analysis of the whole model. No sweep, dihedral or twist were adopted. Table 1 summarizes the values of the final wing geometry.

Figure 3 represent the ultimate wing used for the project. To the basic design, flaps and ailerons where added to study the different maneuvers. The dimensions are shown in Table 2.

Parameter	Symbol	Value (Units)
Wing Area	S	134.97 (ft ²)
Aspect Ratio	AR	12.04
Taper Ratio	λ_w	0.7
Span	b	40.31 (ft)
Mean Geometric Chord	MGC	3.35 (ft)
Root chord	C_r	3.94 (ft)
Tip chord	C_t	2.76 (ft)
Thickness-chord ratio	t/c	0.18
Sweep angle	Λ_w	0 (°)

Table 1: Wing measures.

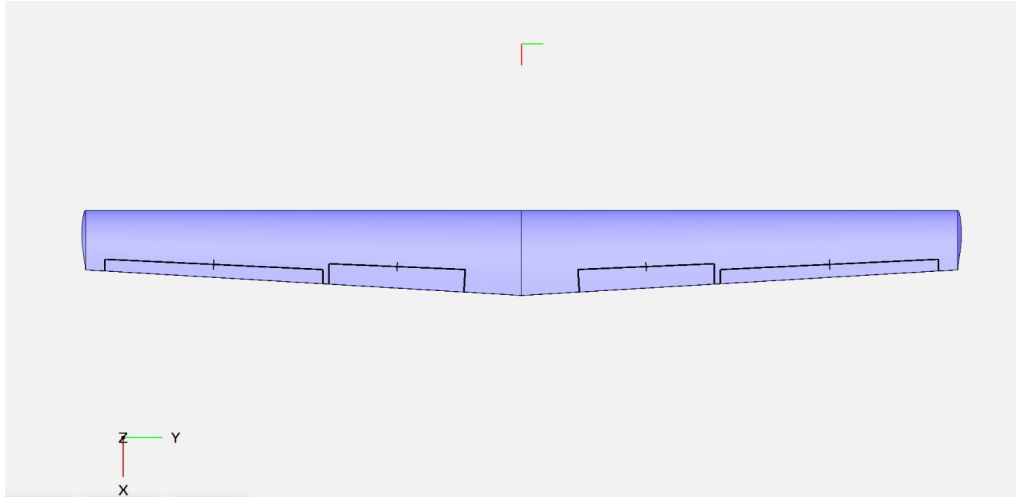


Figure 3: Ultimate wing design.

Element	Length Spanwise (ft)	Length Chordwise (ft)
Flap	6.22	0.952
Aileron	10.08	0.67

Table 2: Flaps and Ailerons dimensions.

2.5 Tail Geometry

For the tail design, an initial horizontal tail volume coefficient, $V_{ht} = 0.7$, and a vertical tail volume coefficient, $V_{vt} = 0.04$, were selected, based on recommendations from Princeton University [6]. With a tail length of $l_t = 9$ ft = 2.74 m (distance from the aerodynamic center of the wing to the aerodynamic center of the tail), it is possible to calculate the vertical tail area S_{vt} , as:

$$S_{vt} = \frac{V_{vt} \cdot S \cdot b}{l_t} = 24.18 \text{ ft}^2 = 2.247 \text{ m}^2 \quad (24)$$

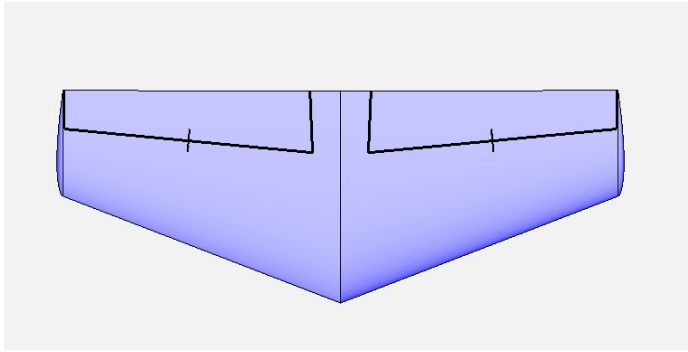
where S is the wing area and b is the wingspan. This ensures sufficient yaw stability. Similarly, the horizontal tail area S_{ht} , is calculated as:

$$S_{ht} = \frac{V_{ht} \cdot S \cdot MGC}{l_t} = 35.15 \text{ ft}^2 = 3.265 \text{ m}^2 \quad (25)$$

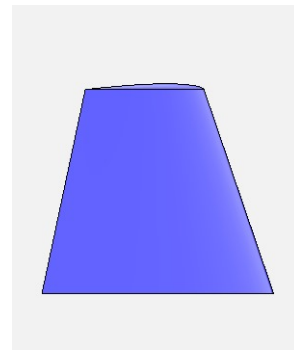
where MGC is the mean geometric chord of the wing. This sizing provides the necessary pitch stability and control for the aircraft.

The aspect ratio for the horizontal tail, $AR_{ht} = 3.5$, was then selected and for the vertical tail, $AR_{vt} = 1.35$, to achieve a balance of aerodynamic efficiency, stability, and structural robustness. The horizontal tail has a moderate aspect ratio of 3.5 and a taper ratio of $\lambda_{ht} = 0.5$ with a straight trailing edge, which provides sufficient lift and stability without the added weight of a higher aspect ratio. For the vertical tail, a lower aspect ratio of 1.35 and a taper ratio of $\lambda_{vt} = 0.6$ with a trapezoidal shape help minimize bending moments from side forces, enhancing yaw stability and reducing structural stress. These configurations ensure that the tail surfaces are effective in controlling the aircraft while remaining durable and lightweight. The tail has a conventional configuration, meaning that the horizontal tail is located at the bottom of the vertical tail.

For both the horizontal and vertical tails, we selected the NACA 0012 airfoil, a symmetrical profile ideal for stability and control. Its zero lift at zero angle of attack provides predictable performance in pitch and yaw, ensuring stable control during maneuvers. The 12% thickness also adds structural strength, making the tail surfaces durable under aerodynamic loads without excessive weight. This profile thus offers an effective balance of stability, control, and robustness for the tail surfaces.



(a) Horizontal Tail



(b) Vertical Tail

Figure 4: Comparison of Horizontal and Vertical Tail Designs

Parameter	Symbol	Value (Units)
Aspect Ratio	AR_{ht}	3.5
Taper Ratio	λ_{ht}	0.5
Span	b_{ht}	11.09 (ft)
Area	S_{ht}	33.14 (ft ²)
Sweep Angle	Λ_{ht}	21 (°)
Mean Geometric Chord	MGC_{ht}	3.169 (ft)

(a) Horizontal Tail Parameters

Parameter	Symbol	Value (Units)
Aspect Ratio	AR_{vt}	1.35
Taper Ratio	λ_{vt}	0.6
Span	b_{vt}	3.81 (ft)
Area	S_{vt}	12.48 (ft ²)
Sweep Angle	Λ_{vt}	18 (°)
Mean Geometric Chord	MGC_{vt}	3.27 (ft)

(b) Vertical Tail Parameters

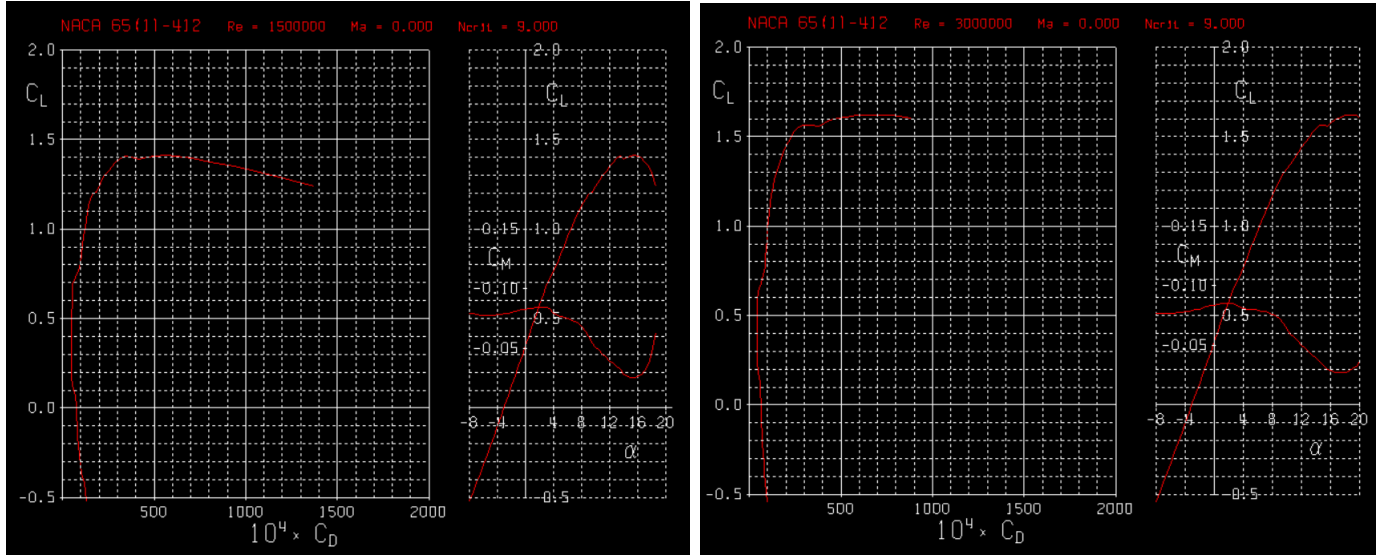
Table 3: Tail Parameters for Horizontal and Vertical Tails

3 Analysis

3.1 Aerodynamics

3.1.1 Sectional Lift Curve and Drag Polar

The NACA 65₁412 aerodynamic analysis was accomplished using XFOil [4] both for stall, Figure 5a, and cruise, Figure 5b.



(a) Airfoil aerodynamic analysis for stall.

(b) Airfoil aerodynamic analysis for cruise.

Figure 5: Airfoil aerodynamic analysis

The drag polar, graph on the left of each image, represents the variation of the lift coefficient based on the drag coefficient expressed in drag counts. The minimum value of the latter both for stall and cruise can be observed on them to which the optimal $C_{l_{stall}}$ and $C_{l_{cruise}}$ correspond.

The lift curves, right one in both figures, give the values of C_l as a function of the angle of attack. From Figure 5a the values of the maximum lift coefficient $C_{l_{max}} = 1.4$ and of the stall angle $\alpha_{stall} = 15^\circ$

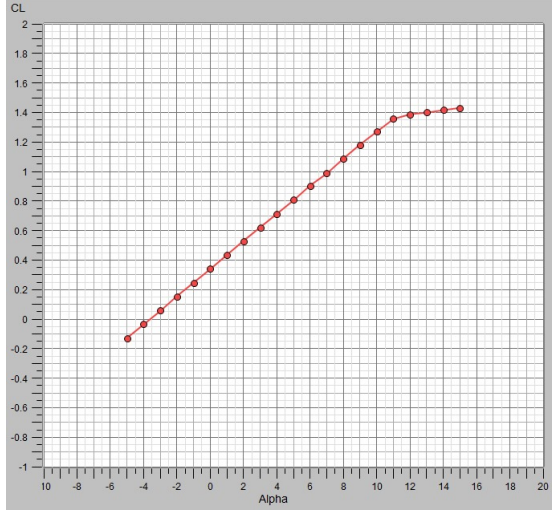
3.1.2 Wing Lift Curve and Drag Polar

The aerodynamic analysis of the wing was carried out using OpenVSP [2] which returned the lift curve, Figure 6a, and the drag polar, Figure 6b.

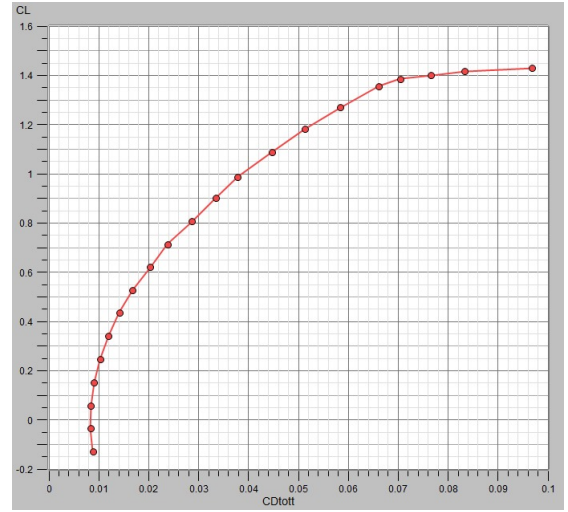
In order to get this results for the C_L over α curve, the 2D drag model of the software was used: the value of the maximum lift coefficient of the airfoil ($C_{l_{max}} = 1.4$) was inserted before running the analysis which returned the following graph based on that value.

The drag polar of the wing was analyzed for an interval of the angle of attack included between -5° and 21° . As it can be deduced from the curve in Figure 6b, the optimal lift coefficient, corresponding to the minimum drag development, is 0.01.

The stall angle of the wing, $\alpha_{stall} = 12^\circ$, was deduced from Figure 6a seeing as it is the angle corresponding to $C_{l_{max}} = 1.4$.



(a) Wing lift curve.



(b) Wing drag polar.

Figure 6: Wing lift curve and drag polar

3.1.3 Drag Analysis

The total drag of an aircraft (Figure 13a) can be divided into three components: the zero-lift-drag, the skin friction drag and the lift induced drag.

$$D = \frac{1}{2} \rho V^2 S_{ref} C_D \approx 67 \text{ lb}_f \text{ (in cruise)} \quad (26)$$

Where

$$C_D = C_{D,b} + C_{D,f} + C_{D,i} = C_{D,0} + C_{D,i} \quad (27)$$

- Zero-Lift Drag Coefficient ($C_{D,0}$): Zero-lift drag is the some of the the basic drag and the zero-lift drag and its value was taken from graph 7b from OpenVSP [2] for α_{cruise} .
- Basic Drag Coefficient ($C_{D,b}$): Basic drag is caused by the pressure distribution over a surface.
- Skin Friction Drag Coefficient ($C_{D,f}$): Skin friction drag is caused by a fluid viscosity as it flows on a surface. It was calculated for the wing with a Reynolds number of 3×10^6 and transition between laminar and turbulent boundary layer was approximated to occur at half the chord. The wing was approximated as a rectangle and the wetted area as twice the wing surface area (S_{ref}): $S_{wet} = 2S_{ref}$. The skin friction coefficient was calculated with Equation 28:

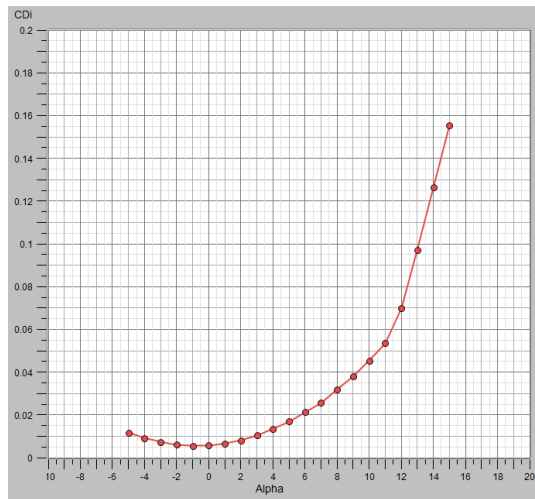
$$C_f = \frac{0.074}{Re^{0.2}} \left(1 - \frac{x_{tr} - x_0}{c} \right)^{0.8} \quad (28)$$

From which the skin friction coefficient was derived using equation 29

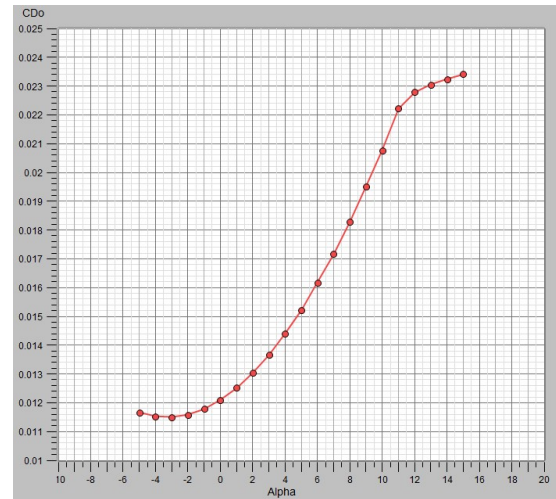
$$C_{D,f} = \frac{S_{wet}}{S_{ref}} C_f \approx 0.0043 \quad (29)$$

- Induced Drag Coefficient ($C_{D,i}$): Induced drag is proportional to the lift that the wing is producing. When the pressure difference of the wing creates wingtip vortices that tilts the

lift vector back thereby creating a backward force relative the motion of the aircraft. Its value was taken from Figure 7a from OpenVSP [2] for α_{cruise} .



(a) Lift induced drag coefficient vs alpha



(b) Zero lift drag vs alpha

Figure 7: Drag analysis

3.1.4 Efficiency

OpenVSP [2] was also used to investigate the efficiency of the whole model. Figure 8 below represents the trend of the lift over drag ratio, equivalent to the maximum gliding distance of the aircraft, depending on different angles of attack.

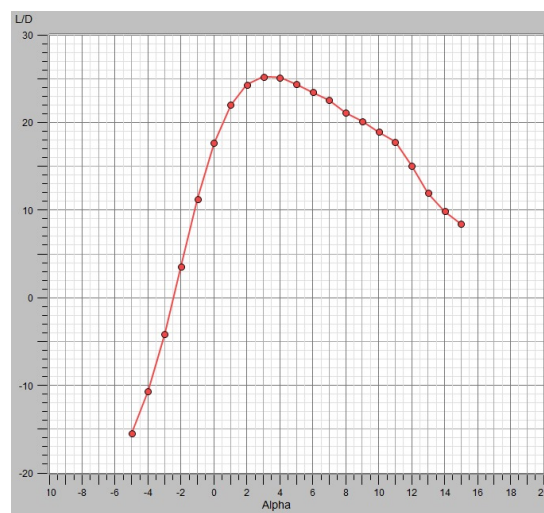


Figure 8: Efficiency of the aircraft.

3.2 Performance

An analysis was conducted to assess the aircraft's performance across various mission stages: range estimation, take-off, climb, cruise, descent, and landing. For each stage, key parameters

such as velocities, take-off and landing distances, gliding distance, and battery requirements are analyzed. The performance evaluation is based on predefined mission requirements mentioned in Section 1.1. For climb and cruise performance Table 4 was constructed showing velocities, dynamic pressure, drag, thrust, climb rate and excess power during climb and cruise [1]. The row highlighted in orange was used for climb, the row highlighted in yellow was used for cruise.

KCAS	KTAS	ft/s	m/s	q [$\frac{\text{slugs}}{\text{ft}^2}$]	D [lbf]	eta_p	T [lbf]	V _v [fpm]	TV-DV [$\frac{\text{ft}\cdot\text{lbf}}{\text{s}}$]
60	71.85	121.27	36.96	11.69	15.78	0.87	260.34	810.98	17571.28
65	77.84	131.37	40.04	13.72	18.52	0.87	240.31	796.78	17263.57
70	83.82	141.48	43.12	15.91	21.48	0.87	223.15	780.22	16904.67
75	89.81	151.58	46.20	18.27	24.66	0.87	208.27	761.11	16490.67
80	95.80	161.69	49.28	20.78	29.68	0.87	195.25	743.66	16145.87
85	101.79	171.80	52.36	23.46	30.72	0.87	183.77	725.05	15842.76
90	107.77	181.90	55.44	26.31	35.50	0.87	173.56	686.70	14878.60
95	113.76	192.01	58.52	29.31	39.56	0.87	164.42	655.60	14204.77
100	119.75	202.11	61.60	32.48	43.83	0.87	156.20	621.05	13456.15

Table 4: Spreadsheet to estimate climb and cruise performance

3.2.1 Endurance and Range

In this section, Endurance and Range are evaluated in order to calculate the battery mass required. First, the range R is calculated:

$$R = EV_{LD_{max}} = 283.14 \text{ km} = 152.88 \text{ NM} \quad (30)$$

where the cruise velocity $V_{LD_{max}}$ is obtained from Section 3.2.4.

The rated capacity of the battery is $C_{rated} = 11000 \text{ Wh}$, from which the mass-specific energy is:

$$E_* = \frac{C_{rated}}{W} = 157.143 \text{ Wh/kg}$$

The LD_{max} has to be calculated as well:

$$LD_{max} = \frac{1}{\sqrt{4kC_{D_{min}}}} = 26.319 \quad (31)$$

where the minimum drag coefficient is $C_{D_{min}} = 0.01$ and $k = \frac{1}{\pi e AR} = 0.03751$. For what concerns the total efficiency: $\eta_{tot} = \eta_{syst}\eta_{prop} = 0.8004$, considering a high propeller efficiency of $\eta_{prop} = 0.87$, given by a variable pitch propeller.

The ratio between the battery mass and the total mass can now be calculated as:

$$\frac{m_{batt}}{m_{tot}} = \frac{R_{LD_{max}}^{9.81}}{3.6\eta_{tot}E_*} = 0.233 \quad (32)$$

which leads to a required battery mass of $m_{batt} = 139.845 \text{ kg}$.

3.2.2 Take-off

The purpose of T-O analysis is to estimate the total T-O distance, its constituent segments, and the associated airspeeds. The results are then compared to the pertinent requirements, answering if the aircraft will meet them. The ground run is the distance from brake release to the initiation of the rotation, when the pilot pulls the control wheel (or stick or yoke) backward to raise the nose of the aircraft.

1. Lift-Off Speed is the speed of the aircraft at which the pilot rotates the airplane to raise from the ground and is calculated as followed:

$$V_{LOF} = 1.1 \sqrt{\frac{2W}{\rho S C_{L_{max}}}} = 42.54 \text{ KCAS} = 50.95 \text{ KTAS} \quad (33)$$

2. The stall speed is calculated using its relation with the Lift-Off Speed as:

$$V_S = \frac{V_{LOF}}{\sqrt{2}} = 38.68 \text{ KCAS} = 46.31 \text{ KTAS} \quad (34)$$

3. The drag and lift coefficients acting on the aircraft during take-off is calculated and ground effects are taken into account according to:

$$C_{L_{IGE}} = \Phi_L \cdot C_{L_{OGE}} \quad C_{D_{IGE}} = \Phi_L^2 \cdot \Phi_D C_{D_{OGE}} \quad (35)$$

Where the subscripts *OGE* means *out-of-ground-effect* and *IGE* means *in-ground-effect*.

These calculations are evaluated in terms of the *separation ratio*, h/b where h is the distance between the aerodynamic center of the c_{MGC} and the ground and b is the wingspan. The quotient h/b was assumed < 0.5 which in turns gives a $\Phi_L = 1.05$ and $C_{L_{OGE}} = 1.35$ which is the $C_{L_{max}}$ of the aircraft. Which gives the lift coefficient during take-off as:

$$C_{L_{IGE}} = 1.05 \cdot 1.35 = 1.42 \quad (36)$$

For the drag the Φ_D was calculated which in turn is used to calculate the drag coefficient:

$$\Phi_D = \frac{33(h/b)^{1.5}}{1 + 33(h/b)^{1.5}} = 0.92 \quad (37)$$

For $C_{D_{OGE}}$ the calculation was done with regards to the drags relationship to C_L and the lift induced drag constant k .

$$C_{D_{OGE}} = k C_{L_{max}}^2 = 0.068 \quad C_{D_{IGE}} = \Phi_L^2 \cdot \Phi_D C_{D_{OGE}} = 0.069 \quad (38)$$

To calculate the lift and drag the reduced rotational speed, V_{ROT} must be calculated and used in the calculations for the lift and drag acting on the aircraft during take-off.

$$V_R = V_{LOF}/\sqrt{2} = 30.08 \text{ KCAS} = 36.02 \text{ KTAS} \quad (39)$$

$$L_{TO} = \frac{1}{2} \rho V_R^2 S C_{L_{IGE}} = 3739.08 \text{ N} = 740.58 \text{ lb}_f \quad (40)$$

$$D_{TO} = \frac{1}{2} \rho V_R^2 S C_{D_{IGE}} = 183.16 \text{ N} = 41.18 \text{ lb}_f \quad (41)$$

4. The thrust during take-off was calculated using the propeller efficiency, $\eta_P = 0.87$, as well as the engines horsepower P_{BHP} for the required energy of the engine during take-off, which is 45 kW the conversion from kW to BHP is done using a known conversion constant of 1.34102:

$$P_{BHP} = 45 \cdot 1.34102 = 60.35 \text{ BHP} \quad (42)$$

$$T_{TO} = \frac{\eta_P (550 \times P_{BHP})}{V_R/\sqrt{2}} = 2112.54 \text{ N} = 474.92 \text{ lb}_f \quad (43)$$

5. The ground run for take-off is calculated, using a ground friction coefficient $\mu = 0.03$:

$$S_{TO} \approx S_G = \frac{V_R^2 W}{2g[T_{TO} - D_{TO} - \mu(W - L_{TO})]} = 110.49 \text{ m} = 362.51 \text{ ft} \quad (44)$$

3.2.3 Climb

An analysis for the climb was done to ensure that the aircraft possess enough power to permit safe operations. For the climb the power required, excess power, drag and lift during climb as well as the best rate-of-climb and best rate-of-climb airspeed was calculated. In Table 4, important values in the orange colored row was extracted for the calculations.

1. For the power requirements the power available was calculated firstly from the handbook of the engine the power available was set to 49.2 kW but in fact it is lower due to the propeller efficiency.

$$P_{AV} = P_{ENG} \cdot \eta_P = 42804 \text{ W} \quad (45)$$

The power required was calculated by subtracting the P_{excess} from the P_{AV} . The excess power was taken from Table 4 and converted into Watt with a conversion factor of 1.35582.

$$P_{REQ} = P_{AV} - P_{excess} = 42804 - 13945.87 \cdot 1.35582 = 23997.39 \text{ W} = 24 \text{ kW} \quad (46)$$

2. For the drag and lift analysis of the climb the drag was set equal to the thrust which was taken from Table 4. The lift was also calculated using values extracted from Table 4 and equals the weight of the aircraft.

$$D = T = 224.46 \text{ N} = 50.46 \text{ lb}_f \quad (47)$$

$$L = qSC_{L_{climb}} = 1322.772 \text{ lbf} = W_{tot} \quad (48)$$

3. The best rate-of-climb for the aircraft was also extracted from Table 4 as:

$$V_{v_{opt}} = 629.19 \text{ fpm} = 3.20 \text{ m} \cdot \text{s}^{-1} \quad (49)$$

4. The best rate-of-climb airspeed is calculated using the adjusted drag model:

$$V_y = \sqrt{\frac{2}{\rho} \left(\frac{W}{S} \right) \frac{\sqrt{k}}{\sqrt{3C_{D_{min}} + 4kC_{L_{minD}} - \sqrt{k}C_{L_{minD}}}}} = 31.90 \text{ m} \cdot \text{s}^{-1} = 104.64 \text{ ft} \cdot \text{s}^{-1} \quad (50)$$

3.2.4 Cruise

For the cruise analysis you evaluate the aircraft's performance in straight and level steady flight including estimations of optimal airspeeds.

1. The first velocity analyzed is the maneuvering stall speed which is done using a load factor, n . While performing a maneuver the aircraft is subjected to an increased load factor, in this case it was calculated to be 2, from the V-n diagram (Section 3.2.7).

$$V_{S_n} = V_S \sqrt{n} = 54.70 \text{ KCAS} = 65.50 \text{ KTAS} \quad (51)$$

2. The cruise velocity is calculated from the following equation:

$$V_{LD_{max}} = \sqrt{\frac{2}{\rho} \left(\frac{W}{S} \right) \sqrt{\frac{k}{C_{D_{min}}}}} = 85.112 \text{ KCAS} = 101.920 \text{ KTAS} \quad (52)$$

Which is in range of the cruise requirements mentioned in Section 1.1.

3.2.5 Descent

For the descent analysis the focus is to analyze the glide range, best glide speed, minimum angle of descent as well as the sink rate from the cruising altitude of 12000 ft. The equations make use of the lifting and drag coefficients obtained through the OpenVSP software [2].

$$C_{L_{descent}} = 0.176 \quad C_{D_{descent}} = 0.01755$$

1. The maximum gliding distance is the horizontal distance traveled in still air, in nautical miles, per 1000 ft of altitude lost in a glide. It is calculated using:

$$R_{glide} = h \frac{C_{L_{descent}}}{C_{D_{descent}}} = 19.80 \text{ NM} = 36.68 \text{ km} \quad (53)$$

2. Best glide speed is the airspeed for which the aircraft will achieve its maximum range in glide and is calculated with:

$$V_{BG} = \sqrt{\frac{2}{\rho_{cruise}} \left(\frac{W}{S} \right)} \sqrt{\frac{k}{C_{D_{min}}}} = 74.89 \text{ KCAS} = 89.68 \text{ KTAS} \quad (54)$$

3. The minimum angle of descent is the flight path angle to the horizontal is calculated using:

$$\gamma = \arctan \left(\frac{L}{D} \right)_{max} = 0.038 \text{ rad} = 2.18^\circ \quad (55)$$

4. The sink rate is the rate at which an aircraft descends. It was calculated from straight and level sink according to:

$$V_V = \frac{C_{D_{descent}}}{C_{L_{descent}}^{3/2}} \sqrt{\frac{2}{\rho} \left(\frac{W}{S} \right)} = 5.19 \text{ m} \cdot \text{s}^{-1} \quad (56)$$

3.2.6 Landing

The analysis of the landing sequence involved calculating key parameters and metrics necessary for evaluating the landing performance. The parameters include the lift-off airspeed (V_{LOF}) from Equation 33, approach angle (γ_{app}), brake horsepower during landing (P_{bhp}), the obstacle height (h_{obs}), and the wing and tail surface area (S), as well as the aerodynamic coefficients for lift and drag in the landing configuration.

The parameters used for these calculations are:

$$V_{LOF} = 50.95 \text{ KTAS} = 42.54 \text{ KCAS}, \quad \gamma_{app} = 3^\circ,$$

$$P_{bhp} = 26.82 \text{ BHP}, \quad h_{obs} = 50 \text{ ft}, \quad S = 135.0 \text{ ft}^2$$

The values for C_L and C_D was simulated using OpenVSP with flaps at 20° and an α -trim of 1.2. The values used are:

$$C_{L_{landing}} = 0.57, \quad C_{D_{landing}} = 0.0265$$

The velocities during the landing sequence is calculated using the following steps:

1. The stall speed with no flaps is assumed to be:

$$V_{s0} = \frac{V_{LOF}}{1.1} = 46.31 \text{ KTAS} = 38.68 \text{ KCAS} \quad (57)$$

2. Using the approach angle and stall speed with no flaps, the height and distance achieved in the landing flare are determined by:

$$h_F = 0.151 \cdot V_{s0}^2 \cdot (1 - \cos(\gamma_{app})) = 0.3859 \text{ ft} \quad (58)$$

$$S_F = 0.151 \cdot V_{s0}^2 \cdot \sin(\gamma_{app}) = 14.74 \text{ ft} \quad (59)$$

3. The distance covered in the landing approach is determined, using the obstacle height and approach angle, by:

$$S_A = \frac{h_{obs} - h_F}{\tan(\gamma_{app})} = 946.7 \text{ ft} \quad (60)$$

4. The airspeed at touchdown (V_{TD}) is assumed to be equal to the airspeed at lift-off (V_{LOF}), calculated in Equation 33, distance traveled during free roll (S_{FR}), and airspeed at which braking is first applied (V_{BR}) are determined by:

$$V_{TD} = V_{BR} = V_{LOF} = 50.95 \text{ KTAS} = 42.54 \text{ KCAS}, \quad S_{FR} = |V_{TD}| = 85.99 \text{ ft}, \quad (61)$$

5. The effective airspeed at which braking is first applied is determined by:

$$V_{BR_{eff}} = \frac{V_{BR}}{\sqrt{2}} = 36.02 \text{ KTAS} = 30.08 \text{ KCAS} \quad (62)$$

Using the values above, the breaking distance is calculated using the following steps:

1. Using the airspeed during braking and the landing configuration, the lift and drag forces acting on the aircraft while braking are calculated by:

$$L_{landing} = 0.25 \cdot 1,225 \cdot V_{BR}^2 \cdot C_{L_{landing}} \cdot S = 338.0 \text{ lb}_f \quad (63)$$

$$D_{landing} = 0.25 \cdot 1,225 \cdot V_{BR}^2 \cdot C_{D_{landing}} \cdot S = 15.71 \text{ lb}_f \quad (64)$$

2. Using the braking force and aircraft weight, the braking distance is calculated in imperial units using:

$$S_{BR} = \frac{V_{BR}^2 \cdot W}{2 \cdot 32.174 \left(\frac{\sqrt{2} \cdot \eta_p \cdot 550 \cdot P_{bhp}}{V_{BR}} - D_{landing} - \mu \cdot (W - L_{landing}) \right)} = 916.6 \text{ ft} \quad (65)$$

3.2.7 Load Factors (V-n diagram)

The V-n diagram, also called flight envelope, is shown in Figure 9. This diagram depicts the capability of the designed aircraft for what concerns minimum and maximum airspeed as function of altitude. From the diagram it is possible to highlight some important parameters, such as the maximum positive load factor ($n = 4$), the maximum positive and negative speeds and the dive speed, all labeled in the figure.

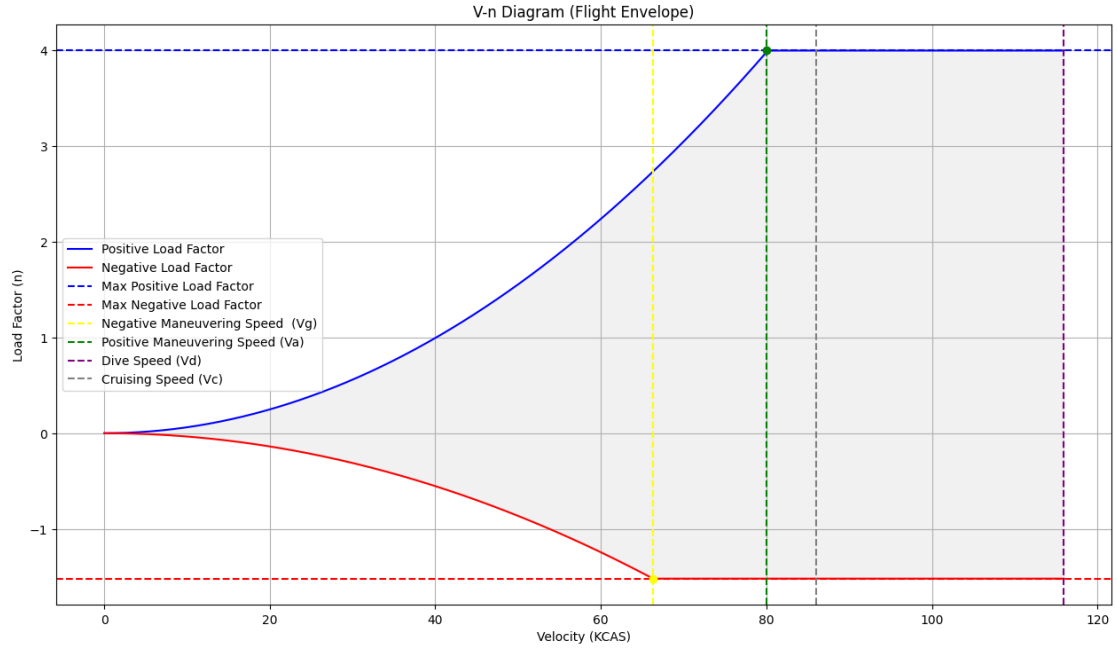


Figure 9: V-n diagram

Parameter (Symbol)	Value (Units)	Parameter (Symbol)	Value (Units)
Location of ac (x_{ac})	0.6564 (ft)	Fractional distance cg (h_{cg})	0
Location of cg (x_{cg})	0	Tail arm (l_{HTcg})	8.1193 (ft)
MGC (c_{mgc})	3.35 (ft)	AOA at 0 Lift (α_{zl})	-1.5 (deg)
Fractional distance ac (h_{ac})	0.25	Wing angle-of-incidence (i_W)	0 (deg)
Ratio of 2-d lift curve W (κ_W)	0.0154	HT efficiency factor (η_{HT})	0.963
Ratio of 2-d lift curve HT (κ_{HT})	0.25	HT angle-of-incidence (i_{HT})	-3 (deg)
Residual downwash angle (ϵ_0)	0.0077 (rad)	Neutral point (h_{NP})	1.140 (ft)

Table 5: Wing and Tail Parameters

3.3 Longitudinal Stability and Control

3.3.1 General Parameters

The lift coefficient for the wing and tail is given by:

$$C_{L\alpha_W} = \frac{C_{l\alpha_W}}{1 + \frac{C_{l\alpha}}{\pi AR}} = 0.0968 \quad C_{L\alpha_{HT}} = \frac{C_{l\alpha_{HT}}}{1 + \frac{C_{l\alpha}}{\pi AR_{HT}}} = 0.128 \quad (66)$$

These equations calculate the lift coefficients for the wing ($C_{L\alpha_W}$) and tail ($C_{L\alpha_{HT}}$) considering their aspect ratios (AR for wing, AR_{HT} for tail). A higher aspect ratio improves aerodynamic efficiency by reducing induced drag, making the glider more efficient in flight.

The total wing and tail lift coefficients are:

$$C_{L_W} = C_{L_{0W}} + C_{L\alpha_W} \cdot \alpha_W \quad \text{and} \quad C_{L_{HT}} = C_{L\alpha_{HT}} \cdot \alpha_{HT} \quad (67)$$

The lift coefficient depends on the angle of attack (α). By calculating C_{L_W} and $C_{L_{HT}}$, we determine how much lift the wing and tail generate at specific angles of attack. This is crucial for balancing the aircraft in flight.

3.3.2 Moment Coefficients

The residual pitching moment and the longitudinal stability derivative are given by:

$$C_{M_0} = C_{M_{0W}} + C_{M_{0HT}} = 0.149 \quad (68)$$

$$C_{M_\alpha} = C_{M_{\alpha W}} + C_{M_{\alpha HT}} = -0.110 \quad (69)$$

The moment coefficients for the wing and tail are given by:

$$C_{M_{0W}} = C_{M_{acW}} + C_{L_{0W}} \cdot (h_{cg} - h_{ac}) = -0.111 \quad (70)$$

where h_{cg} and h_{ac} are defined as the ratio between $x_{cg}/\bar{c}$ and $x_{ac}/\bar{c}$.

$$C_{M_{\alpha W}} = C_{L_{\alpha W}} \cdot (h_{cg} - h_{ac}) = -0.0242 \quad (71)$$

$$C_{M_{0HT}} = \eta_{HT} \cdot V_{HT} \cdot C_{L_{\alpha HT}} \cdot (\epsilon_0 + i_W - i_{HT}) = 0.260 \quad (72)$$

$$C_{M_{\alpha HT}} = -\eta_{HT} \cdot V_{HT} \cdot C_{L_{\alpha W}} \cdot (1 - \delta\alpha) = -0.0862 \quad \eta_{HT} = \frac{1}{1 + w/V_{cruise}} = 0.976 \quad (73)$$

These equations compute the moments generated by the wing and tail based on their respective lift coefficients and positions relative to the center of gravity (CG) and aerodynamic center (AC). The moments influence the stability of the glider by affecting how the aircraft pitches in response to aerodynamic forces.

The total moment coefficient is:

$$C_M = C_{M_0} + C_{M_\alpha} \alpha + C_{M_{\delta_e}} \delta_e + C_{M_{i_{HT}}} i_{HT} \quad (74)$$

The total moment coefficient is a key indicator of the aircraft's stability, and depends on the angle of attack. By summing the individual contributions from the wing and tail, we can ensure that the aircraft achieves near-zero pitching moments for stable, controlled flight.

3.3.3 Neutral Point and Stability

The neutral point is calculated as:

$$h_{NP} = h_{ac} + \frac{n_{HT} V_{HT} C_{L_{\alpha HT}}}{C_{L_{\alpha W}}} \left(1 - \frac{d\epsilon}{d\alpha} \right) = 0.347 \text{ m} \quad (75)$$

The neutral point is critical for assessing the aircraft's static stability. To maintain stable flight, the CG should be positioned ahead of the neutral point. This ensures that the aircraft naturally returns to equilibrium when disturbed.

The center of gravity (CG) and aerodynamic center (AC) were determined using the following relations:

$$\bar{x}_{ac} = \frac{\bar{x}_{ac_w} + \frac{C_{L_{\alpha h}}}{C_{L_{\alpha wf}}} \eta \frac{S_h}{S} \bar{x}_{ac_h} \left(1 - \frac{d\epsilon}{d\alpha} \right)}{1 + \frac{C_{L_{\alpha h}}}{C_{L_{\alpha wf}}} \eta \frac{S_h}{S} \left(1 - \frac{d\epsilon}{d\alpha} \right)} = 0.258 \text{ m} \quad (76)$$

$$SM = h_{ac} - h_{cg} = 25(\%) \quad (77)$$

Equation 77 defines the static margin (SM), which is a key parameter in determining the longitudinal stability of the aircraft. A positive static margin (where $\bar{x}_{ac}$ is behind $\bar{x}_{cg}$) indicates that the aircraft is statically stable, meaning it will naturally return to equilibrium after a disturbance. The larger the static margin, the more stable the aircraft will be, but if it's too large, it may reduce maneuverability.

3.3.4 Trim Conditions

The trim condition ensures that the aircraft can fly straight without requiring control input. For an electric general aviation aircraft, these conditions include achieving steady flight at a specific trim velocity and angle of attack.

$$C_{L\alpha} = C_{L\alpha W_F} + \eta_{HT} C_{L\alpha_{HT}} \frac{S_H}{S} \left(1 - \frac{d\epsilon}{d\alpha} \right) = 0.129 \quad (78)$$

This equation ensures that the lift generated by the wing and tail is balanced, achieving a level flight attitude without any control inputs. This is essential for steady, efficient flight.

The trim velocity (V_{trim}) is given by:

$$V_{\text{trim}} = \sqrt{\frac{2W}{C_{L_{\text{trim}}} \rho S}} = 53.083 \text{ m} \cdot \text{s}^{-1} \quad (79)$$

Where: W is the weight of the aircraft, $C_{L_{\text{trim}}}$ is the lift coefficient at trim conditions, ρ is the air density, S is the wing area.

This equation calculates the trim velocity, which is the speed at which the aircraft generates sufficient lift to exactly balance its weight while maintaining a level flight condition. Achieving this velocity ensures the aircraft can fly without requiring continuous adjustments. This is also a method that allows the confirmation of the expected cruise speed.

The trim angle of attack without elevators deflection (α_{trim}) is:

$$\alpha_{\text{trim}} = -\frac{C_{M_0}}{C_{M_\alpha}} \quad (80)$$

The lift coefficient to trim is

$$C_{L_{\text{trim}}} = C_{L_\alpha} \alpha_{\text{trim}} + C_{L_{\delta_e}} \delta_{\text{trim}} \quad (81)$$

We can use this equation to obtain the trim angle of attack considering the deflection of the elevator [7]:

$$\alpha_{\text{trim}} = \frac{C_{L_{\text{trim}}} - C_{L_{\delta_e}} \delta_{\text{trim}}}{C_{L_\alpha}} \quad (82)$$

The following equation for the elevator angle to trim:

$$\delta_{\text{trim}} = -\frac{C_{M_0} C_{L_\alpha} + C_{M_\alpha} C_{L_{\text{trim}}}}{C_{M_{\delta_e}} C_{L_\alpha} - C_{M_\alpha} C_{L_{\delta_e}}} \quad (83)$$

Where: C_{M_0} is the moment coefficient at zero lift, C_{M_α} is the slope of the moment coefficient with respect to angle of attack.

This equation determines the angle at which the pitching moment becomes zero, meaning the forces acting on the aircraft are balanced. At this angle of attack, the aircraft is in a steady flight

condition with no additional input needed to maintain level flight. Without considering any deflection of the elevator, it was found that $\alpha_{trim} = 1.352^\circ$. With an elevator deflection of $\delta_e = -5^\circ$, $\alpha_{trim} = 5.634^\circ$, as shown in Figure 10.

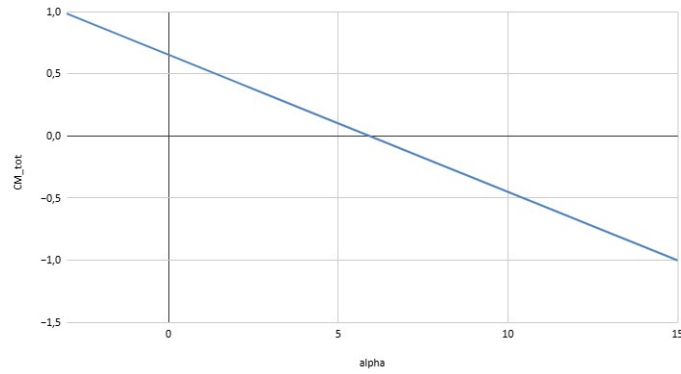


Figure 10: Total pitching moment coefficient (elevator deflected)

4 Results

4.1 Aircraft Configuration

The final configuration adopted is represented in Figure 11.

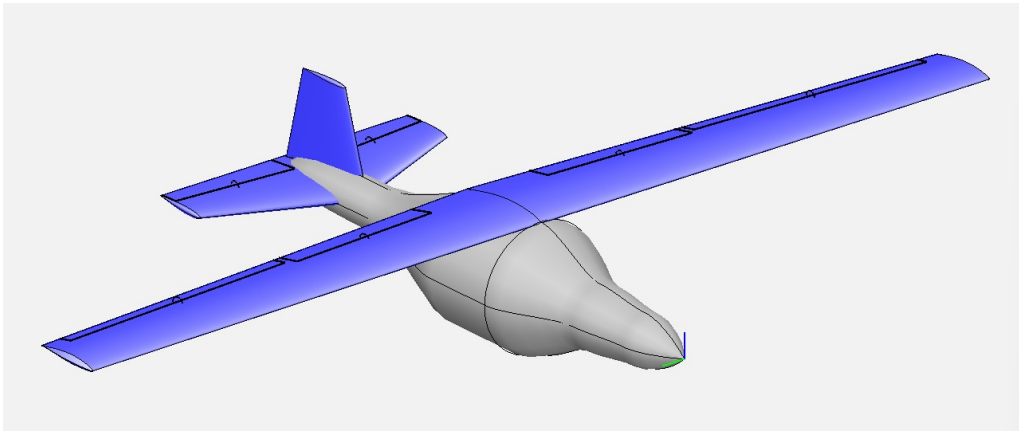


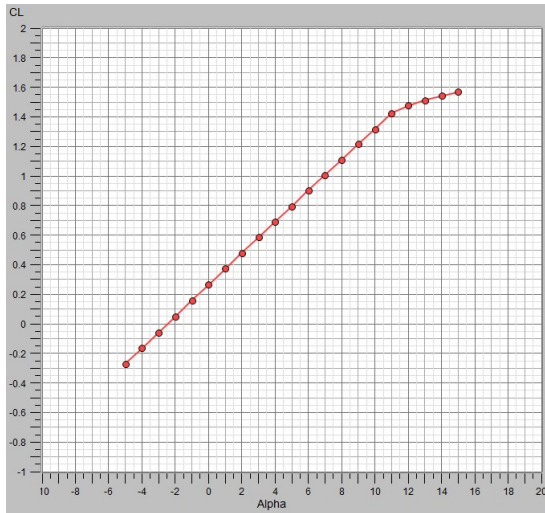
Figure 11: Final aircraft configuration.

4.2 VSPAero Simulation

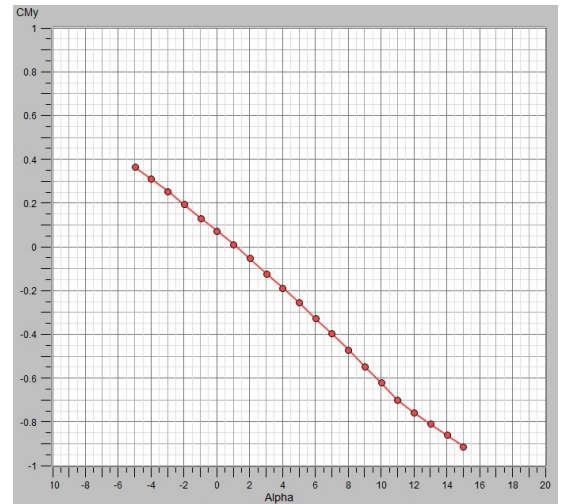
The following graphs were extracted from VSPAero [2] - modelling the main wing, fuselage and tail section. The specific values of Mach number and Reynold's number can be found in Table 6.

Variable	Value
Mach number	0.186
Reynold's number	3×10^6

Table 6: Table with values of the variables used for VSPAero simulation.

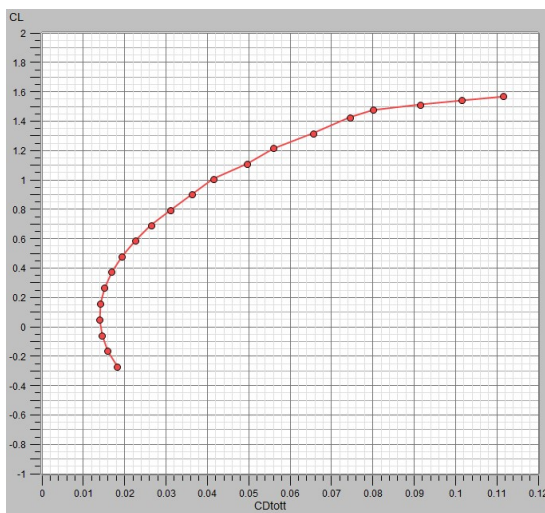


(a) Graph showing lift coefficient vs. angle of attack for the whole aircraft.

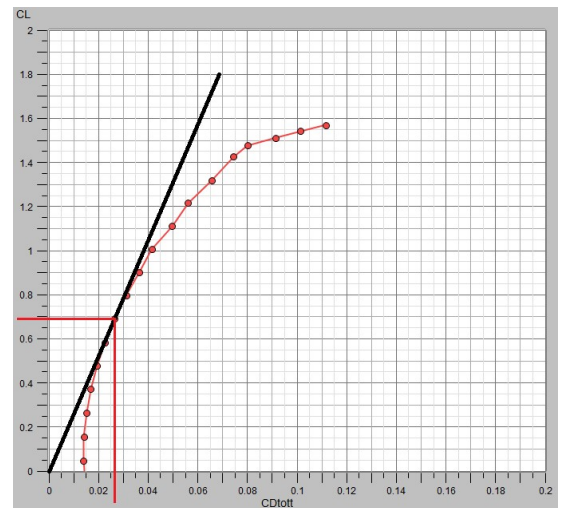


(b) Graph of the pitching moment coefficient vs. angle of attack for the aircraft.

Figure 12

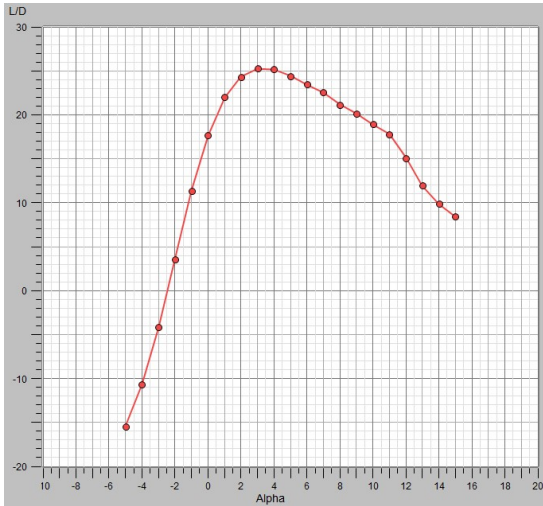


(a) Graph showing the drag polar of the whole aircraft.

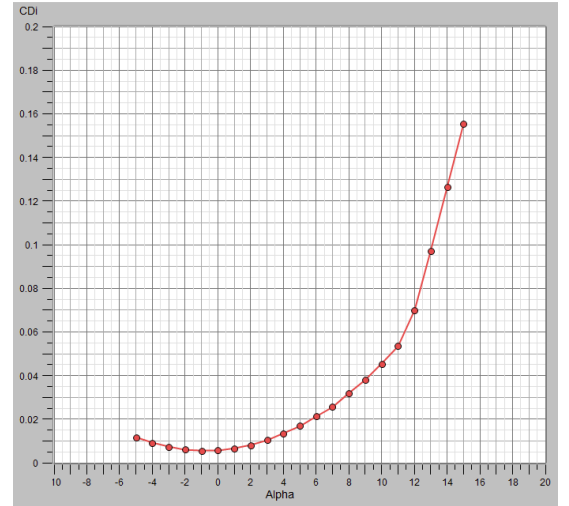


(b) Graph showing the drag polar of the whole aircraft with tangent line.

Figure 13



(a) Graph showing the efficiency vs. angle of attack for the whole aircraft.



(b) Graph showing the induced drag vs. angle of attack for the whole aircraft.

Figure 14

5 Discussion

Generally, with a trainer aircraft, one would strive for a positive C_L at $\alpha = 0$. Figure 12a represents indeed this case. The maximum $C_L = 1.4$ was derived from the airfoil characteristics and the figure does not accurately depict the behavior when going past that point. Hence, stall characteristics of the whole aircraft should not be determined using this data from VSPAero.

Regarding the $C_{My} - \alpha$ - plot, for the aircraft to be statically longitudinally stable, it should have a negative slope and generally a positive pitching moment coefficient at $\alpha = 0$, as clearly showed in Figure 12b. One can discern that when the angle of attack is zero the pitching moment coefficient is positive and approximately 0.075. This means that the aircraft experiences a nose-up pitching moment when the angle of attack is zero. Increasing the angle of attack successively decreases the longitudinal pitching moment until it is zero, which occurs at around 1.2° . This is the trim angle of attack (α_{trim}). At this point, there is no moment about the centre of gravity in steady level flight. Keeping $\alpha_{trim} = 1.2^\circ$ in mind and cross-referencing with Figure 12a one can see that $C_L \approx 0.39$ at $\alpha = 1.2^\circ$. This means that during cruise, with the flow conditions used in the modelling with VSPAero, the aircraft has a lift coefficient of 0.39 and is cruising with an angle of attack of 1.2° .

Figure 14a shows the efficiency of the aircraft, that is the lift over drag, for different angles of attack. For a propeller aircraft, the maximum range can be derived from LD_{max} and ideally one would strive for a α_{trim} which coincides with LD_{max} . For the designed aircraft this is almost the case as α_{trim} is close to its maximum at $\alpha = \alpha_{trim}$. The maximum efficiency instead occurs at $\alpha \approx 3^\circ$. In order for the aircraft to be better optimized - further modelling would be required.

Figure 13a depicts the drag polar for the aircraft and shows the total drag the aircraft produces to maintain a certain lift coefficient. The tangent of a straight line from the origin in this graph would give LD_{max} as $\frac{C_L}{C_D}$, as demonstrated in Figure 13b. Figure 14b shows how the lift induced drag varies with angle of attack: as expected, when the angle of attack increases, so does the lift induced drag.

6 Conclusion

The designed 2-seater General Aviation electric aircraft successfully meets the specified requirements, demonstrating its feasibility and performance.

Key choices, such as selecting the NACA 65₁412 airfoil, provided an excellent lift-to-drag ratio crucial for maximizing range and efficiency. The aircraft achieves a cruise altitude of 12,000 ft with a velocity of 85 KCAS, ensuring compliance with the design parameters. A maximum take-off weight of 600 kg and an empty weight of 419.6 kg highlight the lightweight yet robust configuration, with structural elements carefully optimized using advanced materials.

Performance evaluations confirmed that the aircraft can sustain an endurance of 90 minutes, achieve efficient climb rates, and meet the required take-off and landing distances. Moreover, the calculated trim conditions and aerodynamic analyses validated the stability and control parameters, ensuring safe and efficient operation during all flight phases.

In conclusion, the project demonstrates the potential of electric propulsion in general aviation, combining energy efficiency with sustainability. Future improvements could refine the aerodynamic and structural designs further, extending range and enhancing overall performance.

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Appendix A

The parameters used in the equations of the constraint diagrams are listed in the following table:

Parameter	Symbol
Lift coefficient during T-O run	$C_{L\ TO}$
Drag coefficient during T-O run	$C_{D\ TO}$
Dynamic pressure.	q
Max lift coefficient in T-O config.	$C_{L_{max}}$
Ground run	S_G
Ground friction constant	μ
Acceleration due to gravity	g
Airspeed	V_∞
Vertical speed	V_V
Expected maximum L/D	LD_{max}
Desired climb angle	γ
Minimum drag coefficient	$C_{D_{min}}$
Lift-induced drag constant	k
Load factor	n
Specific excess power at the condition	P_S
Wing area	S
Expected best rate-of-climb airspeed	V_Y
Obstacle height	h_{obst}
Max lift coefficient in the landing configuration	$C_{L_{max}}$
Lift coefficient during ground roll	$C_{L\ LDG}$
Drag coefficient during ground roll	$C_{D\ LDG}$
Time for free roll before braking begins	τ
Thrust loading during ground roll, where T is idle or reverse thrust	T_{gr}/W

Table 7: Table for all parameters in the equations of the constraint diagram

Appendix B

Distances and masses of the components of the whole model for the calculation of the CG.

Part of the a/c	Mass (kg)	Distance from datum (x)	Mass-distance
Fuselage	104.46	0.30	32.32
Landing gear	23.67	0.10	2.37
Furnishing	3.06	0.30	0.92
Windows	5.68	-0.10	-0.57
Flight Control System	10.08	0.01	0.10
Avionics System	1.14	0.01	0.01
Wings	62.67	0.40	21.78
Flaps	15.67	0.80	14.53
Horizontal tail	9.65	2.86	27.64
Vertical tail	7.82	2.90	22.68
Payload	180.00	0.30	54.00
1 battery (forward)	70.00	0.50	35.00
1 battery (aft)	70.00	-1.90	-133.00
Motor+power control	29.70	-2.20	-65.34
Cooling system	5.99	-2.20	-13.19

Table 8: Mass and Distance details for Aircraft Components

Appendix C

Airfoil comparison between NACA 23012, NACA 65₁412 and NLF 0414*F*.

Parameter		Airfoil 1	Airfoil 2	Airfoil 3	SCORE		
Target CL _{max}		1,65	1,65	1,65	Score 0 to 1. 1 is best.		
CL for best ROC		0,7997	0,9657	0,9794			
CL for target cruise, CLC		0,3476	0,569	0,5842			
Airfoil name		23012	NLF0414	65(1)412	1	2	3
Airfoil efficiency ratio (CL _{max} /C _{dmin} , high is best)		296,7	424,3	431,6	0,1	0,9	1
Thickness ratio (high is best)		0,12	0,18	0,12	0,75	1	0,75
STALL	Re for data below	1,5e6	1,5e6	1,5e6			
	Cl for AOA=0 (close to CLC is best)	0,1253	0,33	0,345			
	AOA for Cl=0	-1,25	-3	-3,1	0	0,7	1
	Cl _{max} (highest is best)	1,63	1,6	1,4	1	0,9	0,5
	AOA for Cl _{max}	17	18	14	0,9	1	0,6
	Stall characteristics (A,B or C, A is best)	A	A	A	1	1	1
	Re for data below	3e6	3e6	3e6			
CRUISE	C _{dmin} (lowest is best)	0,00567	0,00449	0,00407	0,3	0,5	1
	Sensitive to surface quality (Y/N, N is better)	N	N	N	1	1	1
	Cl of C _{dmin} (close to cruise is best)	0,3476	0,55	0,3567	1	0,8	0,2
	(Cl/C _d) _{max} (highest is best)	130,2378	125,83501	130,6935	0,9	0,1	1
	CL of (Cl/C _d) _{max} (low Cl is best)	1,2047	0,6254	0,5842	0	0,75	1
	Cruise Cm (low C _m is best) a=2	-0,0091	-0,0777	-0,0838	1	0,1	0,1
CLIMB	Re for data below	3e6	3e6	3e6			
	Drag bucket start at Cl	0,1	0,1	0,15			
	Drag bucket ends at Cl	0,8	0,7	0,75			
	Is CLROC _{max} inside drag bucket?	Y	N	N	1	0	0
	Is CLC inside drag bucket	Y	Y	Y	1	1	1
				Sum:	7,95	8,75	9,15

Figure 15: Comparison table of the airfoils.